Mood Lamp - A natural interaction system based on facial expression recognition

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Abstract

The Mood Lamp is a system that makes use of facial expression to communicate to an RGB color lamp an emotional state, and receiving back a response coherent with the emotional state inferred by changing the color of the light emitted.

1 Introduction

The aim of the systems that simulate natural interactions is to remove the mediation between human and machine typical of classic interfaces. Among the main modes of interaction there are natural speech, gestures, gaze tracking and facial expressions. The latter is particularly relevant because it plays a fundamental role in nonverbal communication between human beings.

Regarding the man-machine interaction, the ability to recognize and synthesize facial expressions allows the machine to gain more communication skills, on the one hand by interpreting the emotions on the face of a subject, and on the other by translating their communicative intent through an output, such as movement, sound response or color change

The Mood Lamp project is a system that uses natural interaction to communicate to a physical object an emotional state and receive back a coherent response. In particular, through your facial expression you communicate your emotional state to an RGB color lamp. The lamp, at this point, the interaction will respond to the interaction by changing the color of the light emitted in accordance with the emotional state inferred.

2 State of the art

As mentioned in the introduction, facial expressions play a very important role in the natural interaction and its analysis has always represented a complex challenge.

They are been under study since 1872, when Charles Darwin published the book *The Expression of the Emotions in Man and Animals*[1], in which he expresses his thesis on the universality of emotions as a result the evolutionary process, and considering facial expressions as a residue of a behavior, according to the principle of the "serviceable habits".

These studies were resumed, about a century later, by Paul Ekman and Wallace Friesen[2] who have demonstrated the existence of a set of basic emotional manifestations, transversal within humanity and independent from the socio-cultural environment.

According to the Facial Action Coding System (FACS)[3], developed by Ekman, the emotional manifestations occur through the activation of a series of facial muscles which are described by the action units (AU). This encoding system allows to describe about 7000 expressions that can be found on a human face by the combination of a set of AUs.

A similar approach, but with different motivation, has been realized by MPEG-4, which identifies a set of Face Animation Parameters (FAP)[4], each of which corresponds to the displacement of a subset of the 84 Feature Points (FP) of the face. These offsets are measured in FAP Units, defined as the distance between the fiducial points of the face.

Regarding the modeling of facial expressions generated by emotions, Ekman identifies six basic emotions as discrete categories, such as: anger, disgust, fear, happiness, sadness and surprise.

In the same years are developed further assumptions opposed to the discrete model proposed by Ekman, among them there's the dimensional one by James Russell, who presents the *circumplex model* [5], that is a one-dimensional circular space on which identify the emotional states of a person. Along the vertical axis of this circle are shown the values of arousal, while on the horizontal one of the values of valence.

Always in 1980 Albert Mehrabian adds a further dimension to Russell's model and creates the Pleasure-Arousal-Dominance (PAD)[6] space.

Further studies were carried out by Mehrabian and Valdez[7] about the effects of colors on the emotions, using the three-dimensional model PAD.

3 Theoretical model

The most challenging part of the project Mood Lamp is the implementation of the system for facial expressions recognition. To do this it is necessary to divide the problem into a finite number of sub problems, and make for each of them a subsystem that operates in a sequential manner with others.

The commonly used approach is to start from a process of face detection, move to one of feature extraction and finally to a recognition phase.

In order for the interaction to takes place in the most natural way possible, the system must necessarily work in real time and it is therefore impossible to use computationally expensive techniques to analyze the expression of a face, it would make the system slow causing a loss of quality in the interaction.

To do this, for the implementation of the first phase (face detection), has been used the algorithm of Viola and Jones, consolidated approach and present in major computer vision libraries such as OpenCV.

Once identified the face within the image, it passes to the extraction of the fiducial points of the face through the use of an Active Shape Model.

The process of the expression recognition, finally, is based on a simple process of linear regression, which starting from the distances between the fiducial points, returns in the output a triple of values of pleasure, arousal and dominance.

3.1 Algorithm of Viola and Jones

The face detection algorithm is based on the work of Viola and Jones [8] in which the images are identified by a set of 3 types of Haar features, shown in Figure 1.

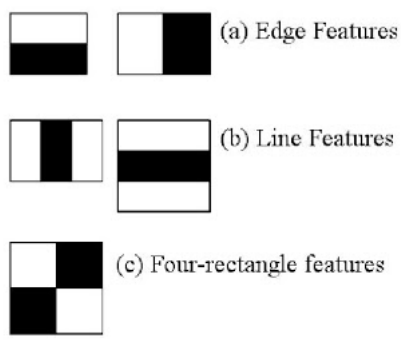

Figure 1: Haar features.

The value of the feature of an image is given by the difference between the sum of the pixels in the white rectangle and the sum of the pixels in the black rectangle. To reduce the computational cost of this operation are used the integral images, in which the value of each pixel is given by the sum of all the previous pixels, formally we have that

$$ii(x, y) = \sum_{x' \leq x, y' \leq y} i(x', y') \tag{1}$$

In this way, in the operations of subtraction is sufficient to consider only one pixel for each rectangle.

Once reduced the image to a set of features, modified version of the algorithm of adaboost is used, that deals to choose the most discriminating feature that separates images of faces from those of non-faces.

The learning algorithm consists of a cascade of classifiers where at each step the number of analyzed features is increased and we proceed to the next step only if you get a positive response from the current classifier.

3.2 Active Shape Model

The Active Shape Models (ASM) [9] are parametric deformable models realized through the use of datasets containing images labeled in a series of control points, called landmarks, on which constraints are defined regarding the eligible deformations.

The evaluation phase is performed automatically by calculating the initial positions of landmarks in the image, and then editing them iteratively varying the parameters within the constraints imposed by the shape, using the value for each landmark which improves the global matching between the model and the image.

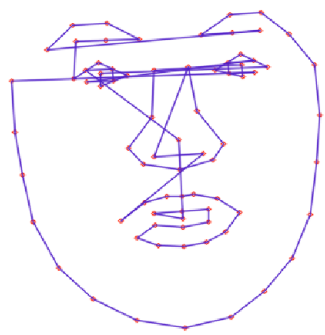

Figure 2: Example of a model of a face using 76 landmarks.

3.3 Linear regression

Regression is a method of supervised learning with output in the continuous space R. This method is used to identify a functional relationship $y = f(x)$ between two values x and y on the basis of the knowledge of them in a finite number of cases, such that y is not too different from actual output value t associated with x.

Not knowing the function f, is assumed to be of linear type, ie that the data are modeled with the straight line

$$f(x; w_0, w_1) = w_1 x + w_0 \tag{2}$$

where x is the independent variable (or regressor), w_0 is the intercept of the regression line and w_1 its slope.

The objective is to minimize a chosen cost function by varying the parameters w_0 and w_1. So, looking for the zeros of the derivatives of the cost function with respect to the two coefficients, we obtain

$$\begin{aligned} \hat{w_0} &= \overline{t_n} - w_1\overline{x} \\ \hat{w_1} &= \frac{\overline{x_n t_n} - \overline{t}\overline{x}}{\overline{x^2} - \overline{x}^2} \end{aligned} \tag{3}$$

where $\overline{t_n}$ and $\overline{x}$ are respectively the arithmetic averages of t_i and x_i of training.

With these values of w_0 and w_1 is possible to make a prediction on the x of test by performing the fitting of the straight line of Eq.(2).

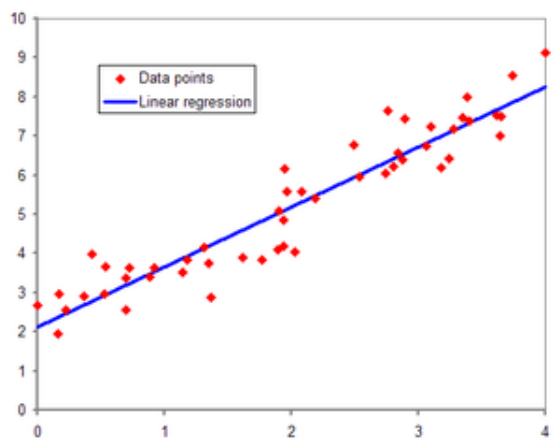

Figure 3: Example of regression line.

In cases where the variable t depends on more than one variable x, the model is said *multiple linear regression* and is expressed in matrix notation as

$$t = Xw$$

where, by multiplying both terms by the matrix X transposed, we obtain

$$X^T t = (X^T X)w$$

that is

$$w = (X^T X)^{-1} X^T t$$

which corresponds to the product of the pseudoinverse of X and the vector t.

4 Simulation and experiments

4.1 Datasets

MUCT http://www.milbo.org/muct/

MUCT[10] dataset consists of 3755 images of faces manually annotated with 76 landmarks corresponding to the fiducial points placed in the contours of the face, mouth, nose, eyes and eyebrows, as shown in Fig. 4. This dataset was used in the project for the construction of the ASM.

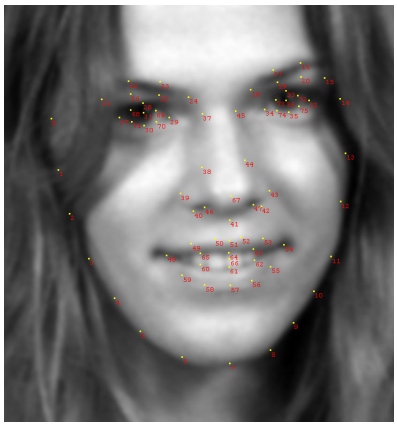

Figure 4: Localization of the 76 landmarks of MUCT dataset.

JAFFE http://www.kasrl.org/jaffe.html

JAFFE[11] dataset consists of 213 images of 10 Japanese women who express 7 facial expressions (6 basic expressions + 1 neutral). A subset of this dataset was used for the training phase of the linear regression.

Figure 5: Example of images in JAFFE dataset.

4.2 System architecture

The project Mood Lamp can be summarized by the framework in Figure 6, it consists of two main components: a subsystem of facial expression recognition (FER) and a subsystem for interfacing with the physical object (the lamp).

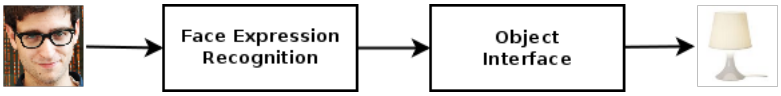

Figure 6: General framework of the system.

Face Expression Recognition System As mentioned in Section 3, the subsystem that deals with the recognition of facial expressions consists of three modules that work in real time and in a sequential manner, summarized in Figure 7

Figure 7: System for the recognition of facial expression.

The first module, the one of Face Detection, takes as input an image captured by the webcam, and if present, takes out the sub-region (ROI) where an human face is recognized.

The subimage is passed to the module that, adapting the model of the fiducial points, is responsible of identifying the locations of the 76 landmarks.

At this point, the module that realizes the regression constructs the features vector by calculating the distances between the fiducial points and performs the fitting with the parameters matrix created during the training phase, returning as output a triple of values $< p, a, d >$ which correspond respectively to the calculated values of pleasure, arousal and dominance, which will be the input of the subsystem that interfaces with the lamp.

Object Interface The subsystem that is responsible for interfacing with the lamp presents no problem in terms of efficiency. This, in fact, consists of two sub-modules, the first of which is responsible to convert in real time the triple of values $< p, a, d >$ received by the previous subsystem in one of values of $< r, g, b >$ corresponding to values of red, green and blue, expressed in the range [0,255].

This triple of values is sent, via the USB port, to the second module, made from a script for Arduino that waits for pending data and at their reception it sets adequately the values of the LEDs contained in the lamp. For completeness is shown in Figure 8 the diagram of this subsystem, which together with the first realizes the comprehensive framework of Figure 6.

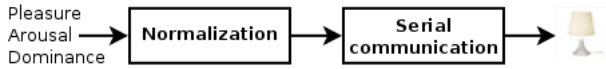

Figure 8: Interfacing system with the lamp.

4.3 Implementative details

Libraries The framework described above has been implemented in C++, while the script to interface with the lamp is in C language and uses the Arduino libraries.

The operations of face detection and image manipulation has been done with the aid of the library OpenCV (http://opencv.org/). For the creation and fitting of the model of the fiducial points was used the library asmlib-opencv (https://code.google.com/p/asmlib-opencv/). Matrix operations in the linear regression, finally, are made using the library Eigen (http://eigen.tuxfamily.org/).

Syntax of the program The software developed provides two main tasks: one of training and on of prediction. For the operation of training are required in input three parameters:

- the file that contains the reference to the training dataset labelled with the values of pleasure, arousal and dominance of each picture (described in Appendix A);
- the output file in which to save the matrix of parameters, the result of regression;
- the file that represents the model to use as ASM for the detection of the fiducial points..

For the operation of prediction, in addition to the parameter that indicates the model to be used as ASM are required:

- the file that contains the matrix of parameters, output of the training phase (described in Appendix B);
- the index of the webcam to use for image acquisitions (default: 0);
- the USB port to which the lamp is connected (default: /dev/ttyUSB0).

Training To realize the labeling of training images with the values of pleasure, arousal and dominance was used a tool called AffectButton (http://mmi.tudelft.nl/ joostb/#AffectButton)[12]. This application allows, by means of an emoticon, to identify the values of PAD that match the expression generated. The values returned by the application have been combined with some of the faces in JAFFE dataset based on the similarity of facial expression produced, as shown in Figure 9.

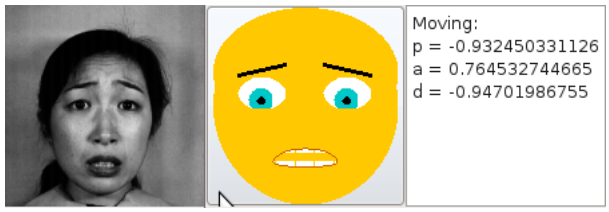

Figure 9: Labelling of the training dataset into PAD space.

Prediction Based on the studies made by Mehrabian and Valdez [7] about the effects of colors on emotions, it was decided a simple correspondence between the values of Pleasure, Arousal and Dominance with those of Red, Green and Blue, according to the equation

$$r = \frac{(p+1)*255}{2}, g = \frac{(d+1)*255}{2}, b = \frac{(a+1)*255}{2} \tag{4}$$

where p, a e d are real values in the range [-1,1], while r, g and b are integer values in the range [0,255].

5 Results

Because the method of ASM fitting is based on tracing of individual points, the position of these points can be inaccurate if the images are noisy, causing erroneous values in the construction of the input array.

During the training phase we see a proper fitting of the model on the images of the dataset, as shown in Figure 10

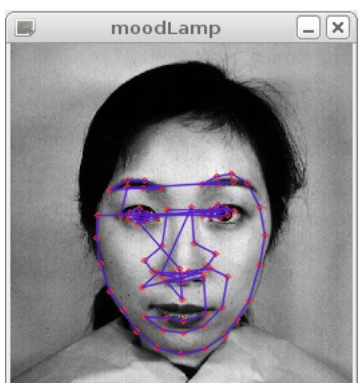

Figure 10: ASM fitting on a training picture.

In the testing phase, however, there are some errors due to the fact that the quality of the light and the webcam used were not optimal. However, considering only acceptable values, we can obtain good results, as shown in Figure 11.

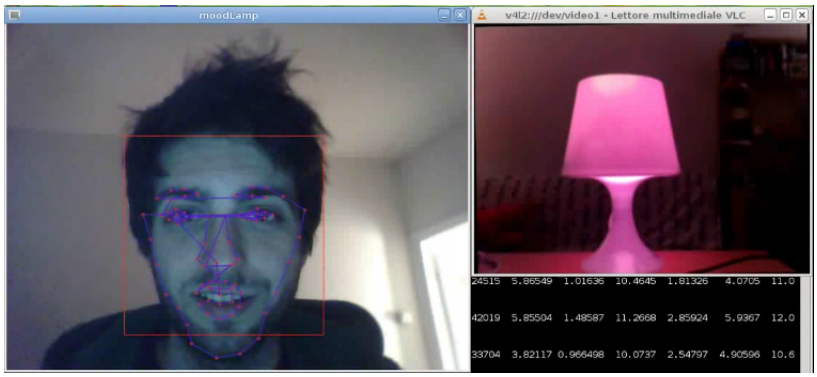

Figure 11: Example of results obtained with the use of the webcam.

A Training file format

```
10              #dataset dimension
01.tiff         #filename
0.0145695364238 #Pleasure value
-1              #Arousal value
0.262251655629  #Dominance value
02.tiff
0.87417218543
0.635025754231
0.917880794702
...
```

B Fitting file format

```
21        #number of rows
3         #number of columns
0         #value w[0,0]
0         #value w[0,1]
0         #value w[0,2]
1.19037   #value w[1,0]
0.40213   #value w[1,1]
1.49484   #value w[1,2]
-0.476484
0.578047
...
```

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